

WiTTA-Bench: Benchmarking Test-Time Adaptation for WiFi Sensing

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Abstract

WiFi sensing offers passive and privacy-preserving perception that complements vision-based sensing, but its performance degrades sharply under domain shifts caused by changes in environment, subjects, or hardware. This challenge is exacerbated in real-world deployments where source data are unavailable, motivating test-time adaptation (TTA) as a practical solution for self-calibration using only unlabeled target samples. We introduce WiTTA-Bench, the first comprehensive benchmark for WiFi TTA, covering 20 representative methods, two adaptation protocols (OTTA and TTDA), and three major physics-induced shifts in WiFi: cross-environment, cross-subject, and cross-device. Furthermore, we contribute a new dataset featuring paired recordings from heterogeneous devices to bridge the cross-device gap. Extensive experiments reveal three key insights unique to WiFi sensing: (i) WiFi domain shifts exhibit a physics-induced hierarchy; (ii) OTTA and TTDA are complementary; (iii) OTTA is generally more robust to hyperparameters, while TTDA is more sensitive due to recursive self-training. WiTTA-Bench establishes the first systematic foundation for adaptive, robust, and deployable WiFi sensing under realistic wireless conditions. Code is available at <https://github.com/BdLI-group/WiTTA-Bench>.

1. Introduction

WiFi sensing is a compelling alternative to vision in camera-restricted settings, including low light, occlusion, and privacy-sensitive environments. Leveraging ubiquitous infrastructure, WiFi provides passive and device-freesensing using *off-the-shelf* hardware, making it attractive for tasks like human activity recognition (HAR). Re-

cent deep models built upon WiFi Channel State Information (CSI) achieve strong *in-domain* performance [10], yet their accuracy collapses under *domain shifts*. WiFi HAR systems are inherently sensitive to physics-induced variations: minor changes in layout, subjects, or hardware distort multipath patterns and can severely degrade performance. This necessitates post-deployment adaptation to ensure robustness in real-world deployments.

Recently, *Test-Time Adaptation* (TTA) [27] has emerged as a promising paradigm that enables a model to self-calibrate at inference using only unlabeled target data. Despite TTA’s remarkable success in computer vision [16], its performance in *wireless sensing* remains largely unexplored. Unlike vision, WiFi CSI domain shifts originate from physical wireless propagation and hardware variations, producing non-stationary and device-dependent distortions that challenge existing TTA algorithms. Moreover, real-world WiFi sensing systems are often deployed on resource-constrained edge devices, requiring lightweight and streaming-friendly adaptation. These constraints expose a fundamental gap: *the field lacks a unified benchmark to systematically evaluate TTA in WiFi sensing*.

To this end, we introduce *WiTTA-Bench*, the first benchmark for evaluating test-time adaptation for WiFi human activity recognition. WiTTA-Bench standardizes datasets, domain splits, and evaluation protocols across both On-line Test-Time Adaptation (OTTA) and Test-Time Domain Adaptation (TTDA). It comprehensively covers three key physics-induced shifts in WiFi sensing: *Cross-Environment* (CE), *Cross-Subject* (CS), and *Cross-Device* (CD). The evaluation integrates 20 representative TTA methods under a consistent framework with comprehensive metrics, including accuracy, latency/adaptation-time, and trainable parameters. Extensive experiments reveal three key insights unique to WiFi TTA: *i) domain shift*: WiFi domain shifts form a clear physics-induced hierarchy: environmental shifts alter multipath statistics, subject shifts deform temporal-spectral geometry, and device shifts restructure

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the entire feature manifold. *ii) methods*: OTTA and TTDA are in complementary deployment regions, i.e., lightweight OTTA is most effective under mild statistical drift, whereas TTDA is needed to realign deep hardware-induced distortions, showing a clear efficiency–adaptability trade-off; and *iii) sensitivity*: OTTA is less sensitive to hyperparameter and scale linear to source-quality, while TTDA is more sensitive due to its reliance on self-training.

WiTTA-Bench provides a reproducible, extensible testbed for studying adaptive, robust, and edge-deployable WiFi sensing, extending the scope of TTA beyond vision.

2. Related Work

WiFi Sensing. WiFi sensing leverages variations in multipath propagation (e.g., RSSI or CSI), to perceive human motion and environmental dynamics. Recent deep models such as DeepFi [33], THAT [10], Person-in-WiFi [37], etc., enable end-to-end spatio-temporal feature learning from CSI, remarkably improving recognition accuracy. However, these models remain highly sensitive to domain shifts (e.g., room layout, subjects, hardware), as CSI distributions can vary drastically in real-world deployments.

Domain Adaptation and Generalization. Domain Adaptation (DA) and Domain Generalization (DG) address domain shifts between training and deployment distributions. Representative methods (e.g., DANN [5], MMD [17], MixStyle [45]) mitigate domain shifts via feature alignment or invariant representation learning, but require source data, hence infeasible for privacy and online use.

Test-Time Adaptation. TTA adapts pretrained models to unseen domains *on-the-fly* using only unlabeled target data without accessing the source domain. Representative methods include BatchNorm recalibration (e.g., TENT [28]), pseudo-label self-training (e.g., SHOT [14]), and consistency-based regularization (e.g., CoTTA [32], EATA [20]). While TTA is popular in computer vision, its application to WiFi sensing remains largely unexplored. A few WiFi-specific efforts exist, e.g., DATTA [25], MDTA [40], and CARING [13]. However, they rely on labeled source domains, domain supervision, or labeled target samples, thus violating the *strict source-free* TTA assumption and making them not directly comparable under our protocol. Our WiTTA-Bench fills this gap by standardizing datasets, protocols, and baselines for the first systematic assessment of TTA in WiFi-based human activity recognition.

3. TTA Settings for WiFi HAR Task

3.1. Preliminary

Problem Definition. Let $\mathcal{D}_S = \{\mathcal{X}_S, \mathcal{Y}_S\}$ denote the labeled source-domain data collected under specific WiFi sensing conditions, and $\mathcal{D}_T = \{\mathcal{X}_T, \mathcal{Y}_T\}$ denote the target-domain data under unseen conditions. Each pair $(\mathbf{x}_i, y_i) \in$

$\mathcal{X}_S \times \mathcal{Y}_S$ follows a probability distribution $P_S(\mathbf{x}, y)$, while target samples at test time t , i.e., $(\mathbf{x}^{(t)}, y^{(t)}) \in \mathcal{X}_T \times \mathcal{Y}_T$ follow $P_T(\mathbf{x}, y)$, where $y^{(t)}$ is unseen for the learner.

A base model $f_{\theta_0}(\cdot)$ is trained on $\mathcal{D}_S$, but may degrade on $\mathbf{x}^{(t)} \sim P_T(\mathbf{x})$ due to $P_T(\mathbf{x}) \neq P_S(\mathbf{x})$. Different from conventional domain adaptation, WiTTA adapts f_{θ_0} *on-the-fly* using unlabeled $\mathbf{x}^{(t)}$ collected at test time, without revisiting or accessing $\mathcal{D}_S$:

$$\theta_t = \arg \min_{\theta} \mathcal{L}_{\text{TTA}}(f_{\theta}(\mathbf{x}^{(t)})), \quad (1)$$

where $\mathcal{L}_{\text{TTA}}$ is an unsupervised adaptation objective (e.g., entropy minimization, pseudo-label consistency, etc.).

Evaluation Metrics. As WiFi HAR is often deployed in resource-constrained environments, besides *accuracy*, we also adapts efficiency metrics: (1) *GFLOPs* per sample for computation, (2) *latency* (ms/sample, OTTA): adaptation + inference time cost per sample; (3) the number of updated *parameters*, and (4) total *adaptation time* (s, TTDA): offline fine-tuning time cost before deployment.

Table 1. **Domain Shifts in WiFi CSI.** Each factor induces distinct signal-level distortions on CSI amplitude and phase.

Shift Type	Source	Impact on CSI
Environment	Room geometry, materials, furniture layout	Changes multipath trajectories and reflection paths, causing delay-spread, path-gain, and phase-coherence variation.
Subject	Body shape, motion style, pose speed	Induces Doppler shifts and dynamic scattering, altering temporal-spectral manifolds.
Device	Antenna pattern, calibration bias, oscillator drift	Causes amplitude scaling, phase offset, and noise-floor mismatch across devices.

3.2. Domain Shifts in WiFi Sensing

Unlike vision, WiFi shifts are *physics-induced*, driven by wireless propagation and hardware effects. They originate from three key factors: *environment*, *subject*, and *device*, which jointly shape multipath propagation [3, 18] and correspond to the key physical perturbations: propagation path, human motion dynamics, and hardware responses (Table 1). Other factors (e.g., transceiver placement or antenna orientation) mainly modulate these mechanisms rather than introduce independent shift dimensions.

Environment. Environmental shifts result from static structures such as room geometry, wall materials, and furniture layout, which alter multipath trajectories and reflection strength [30, 31]. When a room is rearranged or sensing occurs in a new space, dominant paths and delay spread change, modifying amplitude gain and phase coherence. Such variations form the typical CE domain shift.

Subject. Human bodies act as dynamic scatterers. Differences in body shape, posture, or motion speed introduce Doppler shifts and time-varying attenuation [29, 30], producing CS shift even under the same environment.

Device. Hardware diversity introduces further non-stationarity: antenna configuration, calibration bias, oscillator drift, cause amplitude scaling and phase offset [35], creating CD biases that persist even for identical environments and motions.

Table 2. **Standardized TTA Protocols in WiTTA-Bench.** Key differences between online (OTTA) and offline (TTDA) setups.

Attribute	OTTA (Online)	TTDA (Offline)
Source-Free	✓	✓
Streaming Input	✓	✗
Adaptation Timing	Real-time	Pre-deployment
Memory Buffer	✗	Optional
Evaluation	After each batch	After full adaptation

3.3. TTA Protocols

WiTTA-Bench covers two complementary settings: *OTTA* and *TTDA*. Both operate in a source-free manner but differ in when and how adaptation occurs (Table 2).

i) OTTA adapts the model *on-the-fly* from CSI streams using lightweight updates to follow gradual shifts caused by motion or layout changes. OTTA targets real-time adaptation, enabling *streaming and low-latency response*.

ii) TTDA fine-tunes on an unlabeled target dataset typically for several epochs with normalization-only or limited parameter updates, then freezes the model for stable inference. It suits one-shot calibration or edge devices with short offline adaptation windows.

Table 3. **TTA Baselines in WiTTA-Bench.** Representative OTTA and TTDA methods grouped by their core methodologies.

Category	Representative Baselines
A. Online Test-Time Adaptation (OTTA)	
BN Calibration	DUA [19], DELTA [42]
Entropy Minimization	TENT [28], SAR [21]
Pseudo-Labeling	T3A [8], TAST [9]
Anti-Forgetting Regularization	CoTTA [32], EATA [20]
Consistency Learning	RMT [4], PETAL [2]
B. Test-Time Domain Adaptation (TTDA)	
Pseudo-Labeling	SHOT [14], ASL [36], BMD [23]
Consistency Regularization	SFDA-UR [22], APA [26]
Clustering / Alignment	ASFA [34], ISFDA [12]
Source Estimation	BAIT [39], DaC [41]
Self-Supervision	SHOT++ [15]

3.4. Extendable Baselines

We benchmark 20 representative TTA methods (summarized in Table 3) spanning both OTTA and TTDA protocols, covering 5 major methodological categories [16, 43]: normalization calibration, entropy minimization, pseudo-labeling, anti-forgetting regularization, and consistency/self-supervision. All baselines share the same backbone and hyperparameters to ensure fairness.

3.5. Datasets and Implementation

Datasets. We adopt two complementary datasets: *WiHAR-Dual* and *CSLOS* [1]. WiHAR-Dual is *newly collected*

by us to fill the long-standing gap of cross-device evaluation in WiFi sensing. Existing datasets (e.g., MM-Fi [38], Widar3.0 [44]) entangle hardware with other factors and cannot isolate device-induced shifts. WiHAR-Dual addresses this by providing *paired* recordings from two heterogeneous NICs (Intel 5300 and Atheros AR9580) under identical environments, subjects, and activities, enabling the first controlled CD benchmark. CSLOS complements it with occlusion-based LOS/NLOS diversity. The combination of the two datasets enables systematic analysis across three real-world domain shifts. Detailed recording setups and environment layouts are visualized in Appendix §B.

Implementation Details All methods share an identical CNN backbone (four Conv2d-BN-ReLU-MaxPool blocks with [16, 32, 64, 128] channels, followed by a two-layer MLP classifier) and optimization settings for a fair comparison. Full details are provided in Appendix §C.

4. WiTTA-Benchmark

WiTTA-Bench aims to answer three key research questions: **RQ1.** What are the dominant patterns of domain shifts in WiFi sensing? **RQ2.** How effective are TTA methods across shifts? **RQ3.** What factors influence the effectiveness and efficiency of TTA?

Each subsection (§4.1–§4.3) corresponds to one of these questions, investigating the nature of domain shifts, benchmarking adaptation performance, and revealing factors that govern adaptation robustness and efficiency.

4.1. RQ1. Characterizing Domain Shifts

We characterize of domain shifts from both data and model perspectives. At the data level, we directly visualize data distributions of both domains (Fig. 1); at the model level, we examine how a source-trained encoder perceives unseen targets (Fig. 2). These reveal both the physical origins and representational impacts of WiFi domain shifts. The standardized domain splits: CE, CS, and CD, are described in §3.2, with detailed statistics of each domain shown in Table 4. Each domain transfer route (e.g., $E1 \rightarrow E2$) denotes adaptation from a source domain $E1$ to a target domain $E2$.

4.1.1. Feature Shifts

We analyze encoder feature distributions across CE, CS, and CD settings using PCA, t-SNE, and UMAP (Fig. 1) to reveal how CSI representations shift across domains. Full domain shifts analysis can be found in Appendix §D.1.

Cross-Environment (a–b). Source (blue) and target (red) features show clear displacements across environments. The transition from $E1 \rightarrow E2$ is moderate, as both domains are LOS but differ in multipath richness and spatial variability. In contrast, $E1 \rightarrow E3$ shows much stronger separation due to NLOS obstruction, which introduces attenuation, diffraction, and heavier multipath distortion. This

Table 4. **Overview of Standardized Domain Splits in §4.1.** Each protocol isolates one dominant source of domain shift: **CE** for environment, **CS** for subject, and **CD** for device, capturing key variability factors in WiFi-based HAR.

Protocol	Domain ID	Env./Subj./Dev.	Key Attributes	Shift Characterization
A. Cross-Environment (CE)				
CE	E1	Lab Room (LOS)	4.7×4.7 m; TX–RX: 3.7 m; no obstacles	Small enclosed room with stable multipath and high-SNR CSI.
	E2	Corridor (LOS)	7.95×3.6 m; TX–RX: 7.6 m; no obstacles	Long corridor with strong reflections, richer multipath, and larger spatial variability.
	E3	Wooden-Partition Room (NLOS)	Room layout; TX–RX: 5.44 m; 8 cm wooden wall	Obstructed link with attenuation and diffraction; noisy CSI with pronounced multipath distortion.
B. Cross-Subject (CS)				
CS	S1	Subjects (1–6)	Same physical environments as CE	Subject-specific motion; unseen participants in training.
	S2	Subjects (7–10)	Same physical environments as CE	Unseen Subjects with distinct gait and motion patterns.
C. Cross-Device (CD)				
CD	D1	Intel 5300 NICs	30 subcarriers × 3 antennas; standard 802.11n setup	Lower subcarrier resolution.
	D2	Atheros AR9580	56 subcarriers × 1 antenna; 802.11ac-compatible tool	Higher subcarrier resolution and distinct calibration bias.

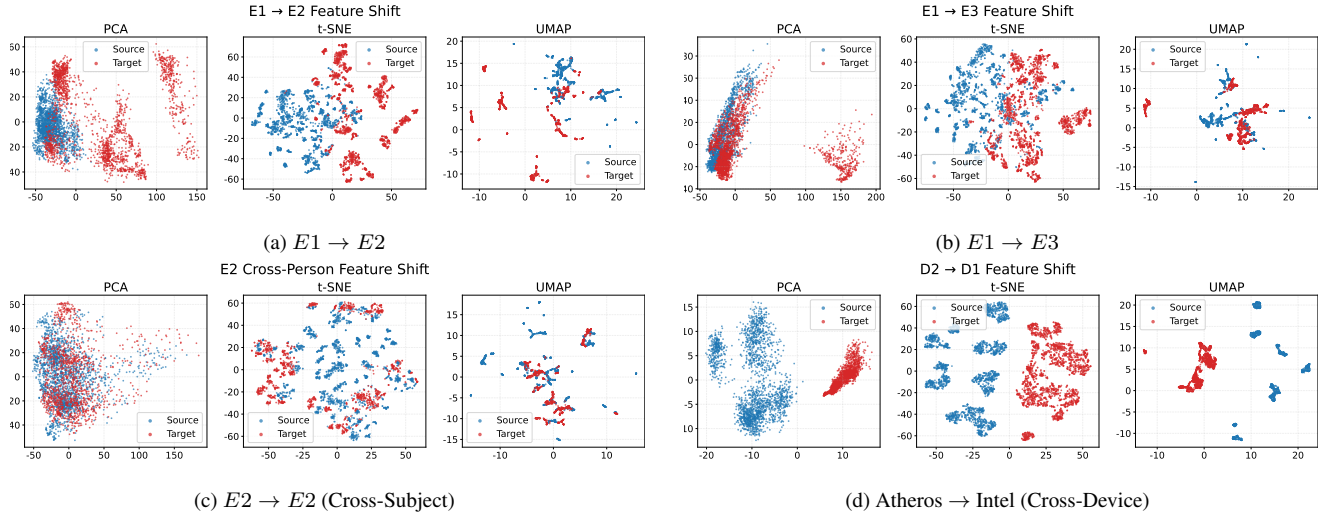


Figure 1. Visualization of **encoder feature distributions** under different WiFi domain shifts. Each map shows PCA, t-SNE, and UMAP embeddings of CSI features from the pretrained source-domain model, where blue and red denote source and target domains. (a–b) Cross-environment (E1–E3), (c) cross-subject (E2), and (d) cross-device (Atheros–Intel) settings. Clear domain gaps highlight the need for robust Test-Time Adaptation under diverse WiFi sensing conditions.

Table 5. **Explicit Domain Shift Metrics on WiHAR-Dual.** Higher Silhouette Score (SS) and Calinski–Harabasz (CH) indicate better cluster separation, while lower Davies–Bouldin (DB) and Entropy imply stronger domain alignment.

Model	Target	SS (↑)	CH (↑)	DB (↓)	Entropy (↓)
M1	E2	0.1054	431.88	2.25	0.1028
M1	E3	0.000021	322.86	2.57	0.1464
M2	E1	0.0202	127.74	2.17	0.2140
M2	E3	0.1207	111.07	2.55	0.2464
M3	E1	0.0367	420.78	2.37	0.2703
M3	E2	0.0424	491.15	2.81	0.3388

order follows the physical layout differences: small-room LOS (E1) → corridor LOS (E2) → NLOS with obstruction (E3), with multipath complexity and signal distortion progressively increasing.

Cross-Subject (c). Compared with the cross-environment case, source and target features show greater global overlap but reorganized internal structures. The t-SNE and UMAP maps show less distinct class boundaries, implying

that intra-class cohesion decreases while inter-class separation weakens. E2 and E3 exhibit the smallest distributional differences, indicating that when the environment remains similar, variations among human subjects mainly lead to geometric perturbations rather than global statistical drifts.

Cross-Device (d). The cross-device setting shows the most pronounced separation. The two domains form nearly disjoint manifolds, and the PCA axes even display mirrored orientations between Atheros→Intel and Intel→Atheros directions. In both t-SNE and UMAP visualizations, the target-domain clusters collapse or flip relative to the source distribution, suggesting that the feature manifolds learned on one hardware platform cannot be directly transferred to another. This reflects device-dependent effects that reshape the representation space.

The visualizations show a progressive domain discrepancies: cross-environment shifts produce moderate statis-

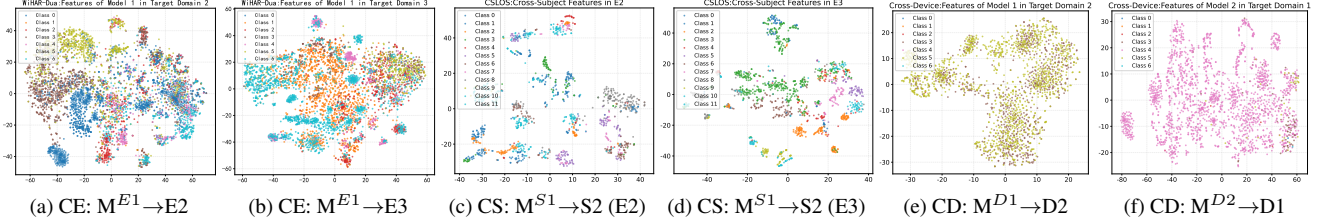


Figure 2. t-SNE visualization of source model features under three WiFi domain-shift settings: CE, CS, and CD. Each color represents a predicted activity class on the target domain. M^{E1} – M^{E3} denote models pretrained on distinct environments, M^{S1} indicates models trained on one group of subjects ($S1$) and adapted to another ($S2$), while M^{D1} and M^{D2} correspond to Intel and Atheros devices, respectively. Feature manifolds show increasing divergence from CE to CD, reflecting progressively stronger physical shifts.

tical displacement, cross-subject shifts introduce local geometric reorganization, and cross-device shifts result in near-complete manifold divergence.

4.1.2. On Class-Level Feature Inconsistency at Test Time

Beyond domain-level feature shift, we further examine how class-level structures behave when the *pretrained model* is exposed to unseen domains. This also in line with the setting of TTA. Fig. 2 visualizes the predicted target-domain embeddings, where each color represents a predicted activity class in t-SNE space. An extended analysis can be found in Appendix §D.2.

From Fig. 2, we can see the degree of class separability varies notably. When the alignment between source and target domains is relatively strong (e.g., $E1 \rightarrow E2$), clusters remain compact and well-separated, forming clear multi-class structures with clear boundaries (Fig. 2 (a)). As the domain gap increases, the embedding geometry gradually becomes irregular: neighboring clusters start to merge, intra-class densities fluctuate, and class boundaries blur (Fig. 2 (b)). In some cases, the overall structure becomes highly entangled, where overlapping or fragmented clusters replace the original organized layout (Fig. 2 (c-d)).

In conclusion, the evolution across settings follows a clear visual trajectory, *compact* $\rightarrow$ *fragmented* $\rightarrow$ *entangled*. Compared with the domain-level observations in the previous section, class-level inconsistency reflects a finer-grained manifestation of the same underlying shifts: statistical displacement at mild domain gaps evolves into geometric deformation and manifold disorganization as the gap widens. This progression highlights how domain shifts disrupt not only the global feature alignment but also the internal class structure learned during source training.

4.1.3. Quantifying Domain Shift with Clustering Metrics

To qualify the domain shift in numerical values, we further compute several clustering-based metrics, including Silhouette Score (SS), Calinski–Harabasz (CH), Davies–Bouldin (DB), and prediction entropy, as summarized in Table 5. Higher SS and CH values imply better compactness and separability of pseudo-labeled clusters, while lower DB and entropy values reflect more confident and discrimina-

tive predictions. For instance, adaptation routes such as $M^{E1} \rightarrow E2$ yield higher SS/CH and lower DB/Entropy, consistent with their compact visual structure in Fig. 2. However, routes like $M^{E1} \rightarrow E3$ show deteriorated clustering scores, aligning with the observed feature dispersion.

4.1.4. Do Domain Shift Metrics Predict TTA Performance?

While the above visualizations and metrics quantify how different domains diverge in feature space, they do not necessarily reveal whether such divergence correlates with the eventual adaptation outcome. In practice, a key question arises: *can these domain-shift metrics serve as a proxy for predicting how well a model will benefit from test-time adaptation?* Addressing this question helps assess whether static geometric indicators, which can be computed before any adaptation, can estimate adaptation difficulty or potential performance gain. Ideally, a strong negative correlation should exist between domain inconsistency and eventual TTA accuracy: domains with greater shifts would benefit more from adaptation.

However, empirical results reveal a weak correlation¹. Additionally experiments using MMD [6] (Appendix §D.3) at both feature and BatchNorm-statistic levels also fail to reliably predict TTA difficulty. As shown in Fig. 2 and Table 5, the ranking of domain pairs based on clustering indices does not align with the actual TTA outcomes in §4.2. For example, $M^{E1} \rightarrow E3$ exhibits the worst clustering scores but not the lowest TTA accuracy, while some moderately misaligned pairs still perform poorly after adaptation. This inconsistency indicates that clustering or distributional metrics such as SS, CH, DB, and MMD only capture local compactness of features and overlook adaptation behaviors, e.g., optimization stability and entropy regularization, which strongly influence TTA performance.

These results highlight a fundamental limitation in WiFi-based adaptation: *observable feature divergence is an imperfect predictor of adaptation difficulty*. Unlike visual domains, WiFi signal distributions are heavily shaped by latent

¹The Spearman correlations r between metrics (Table 5) and post-TTA accuracies (in Appendix §4.2) ranged from -0.21 to 0.18 ($|r| \leq 0.25$).

physical factors (propagation paths, antenna calibration, environmental clutter), causing non-stationary distortions that static representation metrics cannot capture.

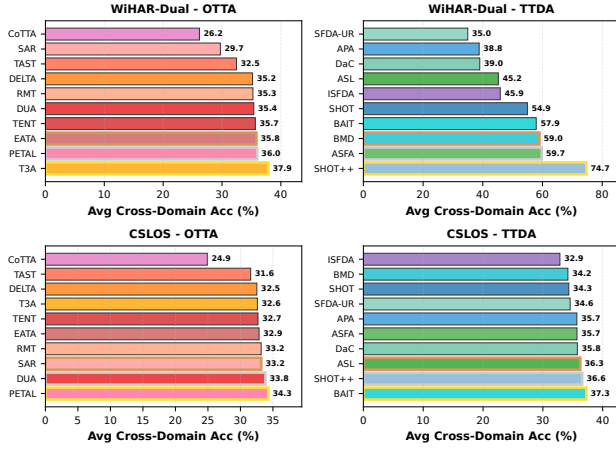


Figure 3. Average cross-domain **accuracy ranking** of representative TTA methods. (a–b) WiHAR-Dual and (c–d) CSLOS under OTTA and TTDA protocols. Lightweight OTTA methods (DUA, EATA, TENT) achieve the best trade-off between accuracy and latency, while clustering- and self-supervised TTDA methods (ASFA, SHOT++) perform best under larger domain gaps.

4.2. RQ2. Benchmarking Test-Time Adaptation

Building upon the defined domain splits, we benchmark representative TTA methods under both OTTA and TTDA protocols. The detailed results are reported in Appendix §D.4 and Tables 7–14.

4.2.1. Overall Performance

Fig. 3 summarizes average cross-domain accuracy rankings across both datasets and adaptation protocols. *lightweight OTTA methods* that calibrate only feature statistics (e.g., DUA, EATA, TENT) achieve the best accuracy–latency trade-off (Fig. 7), while heavier regularization or self-training methods (e.g., CoTTA) provide limited or unstable gains at much higher computational cost. For TTDA, *clustering- and self-supervised* strategies such as ASFA and SHOT++ rank highest on both WiHAR-Dual and CSLOS, demonstrating their ability to reorganize class manifolds under larger domain gaps. By contrast, purely consistency-based or pseudo-labeling approaches (e.g., ASL, DaC) yield moderate but less stable improvements.

Fig. 3 highlights a clear trade-off: *OTTA excels in low-latency statistical calibration*, whereas *TTDA dominates in deeper feature realignment*. This difference aligns with analyses in §4.2.2, which shows WiFi domain shifts can be largely calibrated by simple normalization under on-line constraints, while offline clustering helps correct larger structural distortions.

4.2.2. Discussion: Revisiting Vision vs. WiFi

As revealed by the vision TTA benchmark [43], consistency-based OTTA methods (e.g., CoTTA) usually outperform normalization-based ones (e.g., TENT) when coping with visual style or texture shifts. However, a striking observation from Tables 7–14 is that this trend does not hold for WiFi CSI data. In wireless sensing, domain shifts are dominated by hardware, antenna gain, frequency response, etc., that lead to complex physical distortions rather than semantic or appearance variations. As a result, enforcing prediction consistency consumes considerable computational cost but contributes little to accuracy. Instead, lightweight statistic updates, such as BatchNorm recalibration or entropy minimization, already capture most of the usable distribution drift. In essence, WiFi signals mainly need to be *re-centered* (mean–variance alignment) and *re-confident* (entropy suppression), rather than hallucinated through pseudo-target self-training.

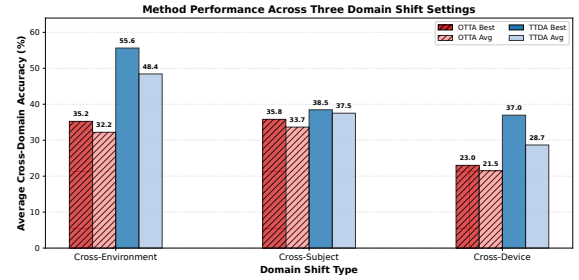


Figure 4. Average performance of representative TTA methods across three WiFi domain shift types. TTDA achieves the highest accuracy under cross-environment shifts, while all methods degrade notably under cross-device shifts.

4.2.3. Breakdown by Shift Type

The detailed accuracy breakdown for all adaptation routes is reported in Tables 7–14. Fig. 4 illustrates the relative difficulty of the three shift types using OTTA (best vs. average) and TTDA (best vs. average) methods. From Fig. 4, we observe a hardness trend, i.e., CE < CS < CD.

Cross-environment. CE remains the most TTA-friendly, as the underlying RF propagation remains similar across rooms, producing a largely statistical, low-rank envelope drift. From Fig. 4, TTDA methods achieve the highest overall accuracy, since clustering- and self-supervised strategies can leverage full target batches to realign the global feature envelope. OTTA methods also yield consistent gains through simple normalization or entropy-based recalibration, but their improvements saturate earlier due to the limited severity of CE drift.

Cross-subject. CS lies in a middle ground. Although each subject exhibits unique gait and body-reflection patterns, their signal manifolds largely preserve a shared topology. Both OTTA and TTDA achieve moderate but consistent improvements, suggesting that person-specific varia-

tion behaves as a *soft geometry shift*, which suggests the feature space changes locally but not structurally.

Cross-device. CD is the most challenging scenario. Transferring between Intel and Atheros cause sharp performance drops for all methods. OTTA yields marginal gains since the RF responses form essentially a *new manifold* tied to hardware-specific channel responses. In contrast, TTDA performs a *structural realignment*, reconstructing the latent geometry of motion in the new RF world and recovering up to +30% absolute accuracy (ASFA). This highlights that device heterogeneity introduces a *hard physics shift* beyond what online normalization can handle.

4.2.4. Revisiting Domain Shifts: Why Cross-Environment is Easier than Cross-Subject?

A counter-intuitive finding is that CE adaptation is easier than CS under TTA, which is contrary to earlier DA observations [3, 11]. The key reason lies in the nature of the shift. CE is dominated by a *global envelope drift* (low-rank mean/variance change). As revealed in Fig. 1, features from the target environment move coherently as a cloud, where the cluster topology remains intact, and only the statistical envelope shifts. In contrast, CS introduces a *motion-manifold reorganization* that alters the intrinsic geometry of the feature space: target samples intrude into and reshape class boundaries, breaking simple re-centering assumptions. Consequently, CE is more *TTA-friendly* for simple statistical calibration, whereas CS requires manifold anchoring or source supervision beyond pure TTA.

4.2.5. Takeaway

OTTA is most effective for *soft, continuous, and statistical* shifts (cross-environment and cross-subject), providing high efficiency and low latency. TTDA is effective for *hard, structural, and discontinuous* shifts (cross-device), where full-feature realignment is required at the cost of offline adaptation time. This observation provides a useful guidance for real-world deployment, i.e., one shall use OTTA for light shifts and TTDA for heavy ones.

4.3. RQ3. Factors Influencing TTA Effectiveness

4.3.1. Parameter Sensitivity

Batch Size and Learning Rate. We visualize representative batch size and learning rate grids in Fig. 5. Comprehensive grids and variance statistics are provided in Appendix §E.1. Normalization-based OTTA methods (e.g., TENT, DUA, EATA) show *broad and flat accuracy trends* across wide hyperparameter ranges, indicating strong robustness to tuning. In contrast, clustering-based TTDA methods (e.g., ASFA) form *sharp, high-gain optima*, where accuracy drops drastically when learning rate or batch size deviates from the optimal point (e.g., ~63% → 35% on WiHAR-Dual). Pseudo-labeling methods such as SHOT remain relatively stable by freezing the backbone and only

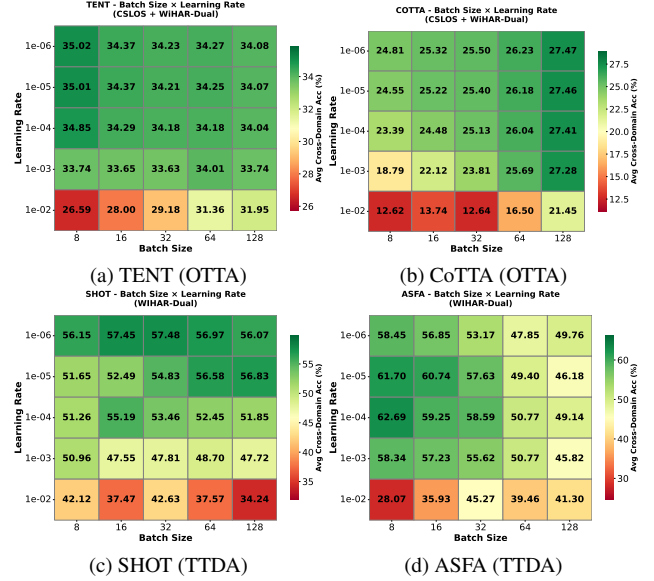


Figure 5. Representative **hyperparameter sensitivity** of OTTA vs. TTDA methods on WiHAR-Dual and CSLOS. Greener shades indicate higher accuracy. OTTA (a,b) exhibits broad stability across learning rate and batch size, while TTDA (c,d) shows higher peaks but sharper optima.

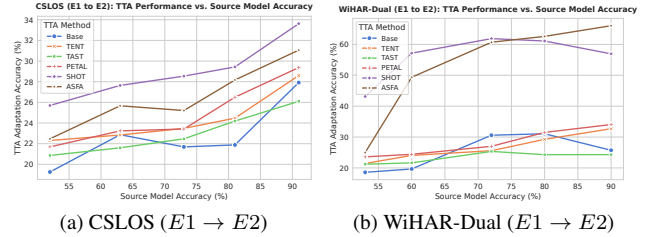


Figure 6. **TTA performance vs. source model accuracy.** Relationship between source model accuracy and downstream TTA performance across two datasets. OTTA methods (e.g., TENT, PETAL) show smoother scalability with increasing source quality, whereas TTDA (e.g., SHOT, ASFA) achieves higher absolute accuracy at high source quality.

adapting the classifier head. These results show that OTTA’s stability originates from its local, low-magnitude statistical updates, while TTDA’s sensitivity arises from recursive pseudo-label feedback.

4.3.2. Initial Model Quality and TTA Gain

To study the effect of source model quality on TTA, we tune source models accuracies from 50% to 100% via early stopping. Fig. 6 shows a clear *positive correlation*: stronger source models consistently yield higher TTA accuracy, though gains gradually saturate. Normalization-based OTTA methods (e.g., TENT, PETAL) present smooth and stable TTA gain with increasing source quality, whereas clustering-based TTDA methods (e.g., SHOT, ASFA) achieve higher peak accuracy but are more sensitive to weaker sources. Interestingly, SHOT shows a slight performance decline at very high source accuracy, indicating

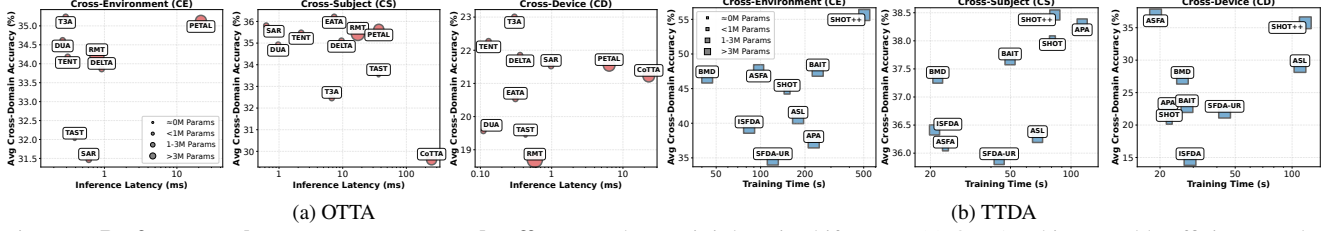


Figure 7. **Performance–latency–parameter trade-offs** across three WiFi domain shift types. (a) OTTA achieves stable efficiency under CE/CS but fails on CD; (b) TTDA achieves higher accuracy under CD at the cost of computation.

that overconfident pseudo-label may induce self-reinforcing bias. Overall, the positive correlation suggests that the *TTA gains are driven by adaptation rather than recovery from source overfitting*.

Table 6. **Backbone generality**. Average accuracy (%) under CE/CS/CD using MobileNetV2 and ResNet-10 on WiHAR-Dual.

Setting	Backbone	Base	TENT	EATA	T3A	SAR
CE	ResNet-10	25.72	39.37	39.87	38.93	39.73
	MobileNetV2	25.10	36.60	36.90	35.30	34.70
CS	ResNet-10	34.77	39.15	39.52	32.28	38.87
	MobileNetV2	34.60	36.60	37.90	33.40	37.50
CD	ResNet-10	14.29	28.55	28.93	14.29	29.02
	MobileNetV2	15.60	39.30	38.50	15.40	38.80

4.3.3. Backbone Generality

To test performance under different backbone, we replace the default CNN encoder with MobileNetV2 [24] and ResNet-10 [7] and evaluate representative OTTA methods under all three shift types on WiHAR-Dual (Table 6). The results show *highly consistent trends* across CE/CS/CD settings (Table 6): normalization-based OTTA remains effective, cross-device adaptation is the most challenging, and method ranking is preserved.

4.3.4. Accuracy-Efficiency Trade-off

Fig. 7 compares the accuracy–efficiency Pareto fronts of representative OTTA and TTDA methods across three WiFi domain shifts. Each bubble denotes one method, with the horizontal axis for latency/adaptation-time, vertical for accuracy, and bubble size for trainable parameters. For OTTA, normalization- and entropy-based methods (e.g., *TENT*, *DUA*, *DELTA*) lie near the upper-left corner, sustaining $\sim 34\text{--}35\%$ accuracy within <10 ms latency and $\leq 1\text{M}$ trainable parameters. Their flat Pareto front reflects a stability-oriented design: lightweight, streaming-friendly adaptation dominated by low-magnitude normalization updates. For TTDA, offline methods (*SHOT++*, *ASFA*, *BMD*) achieve substantially higher accuracies ($45\text{--}60\%$) by trading much longer adaptation time ($80\text{--}500\text{s}$) and more parameters. *ASFA* offers the most consistent Pareto efficiency, reaching $\sim 48\%$ with moderate compute, while *SHOT++* attains the highest peak ($\sim 57\%$) but with more resource cost.

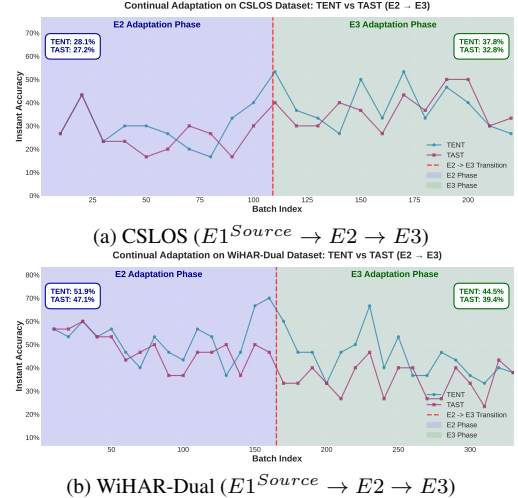


Figure 8. Continual adaptation accuracy for OTTA methods (TENT vs. TAST) on CSLOS and WiHAR-Dual.

4.3.5. Continual Adaptation Analysis

Fig. 8 exemplifies the continual adaptation behavior on CSLOS and WiHAR-Dual. This setting mimics real deployments in which a WiFi system must keep adapting as layouts change or transitions between LOS and NLOS conditions. While CSLOS exhibits smooth statistical transitions across environments, WiHAR-Dual shows stronger oscillations, confirming the route-sensitivity and dataset-dependent dynamics later expanded in Appendix §E.2.

5. Conclusion

We present WiTTA-Bench, the first unified benchmark for TTA in WiFi sensing. It standardizes datasets and protocols to analyze physics-induced shifts across environments, subjects, and devices. Results show that OTTA handles mild statistical drift, whereas TTDA becomes crucial for structural, hardware-driven shifts. While this benchmark focuses on HAR, the CE/CS/CD hierarchy arises from task-agnostic wireless physics (e.g., multipath, scattering, hardware response) and is expected to generalize to other WiFi sensing tasks. We hope the benchmark will simulate more *physics-aware* TTA methods to enable robust and deployable wireless perception.

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